UNIVERSITY OF CAMBRIDGE

Machine Learning and the Physical World

Lecture 6 : Statistical Emulation

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28th October, 2026

http://carlhenrik.com

Taking stock

- Why is uncertainty not just desirable but necessary for learning

Taking stock

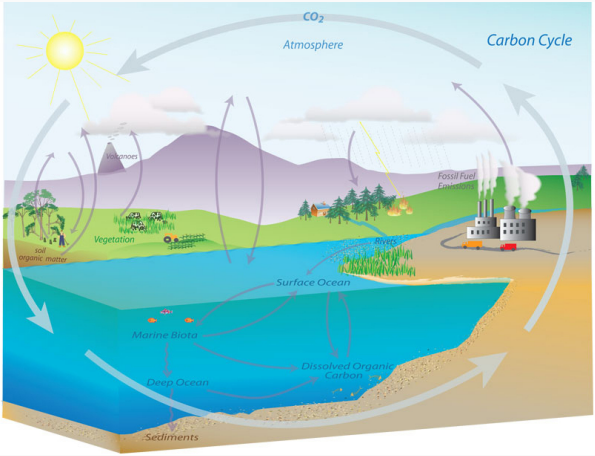
- Why is uncertainty not just desirable but necessary for learning
- Why any prediction can only be interpreted in the light of the beliefs that lead to them

Taking stock

- Why is uncertainty not just desirable but necessary for learning
- Why any prediction can only be interpreted in the light of the beliefs that lead to them
- Mathematical formalism for computing with uncertainties

Taking stock

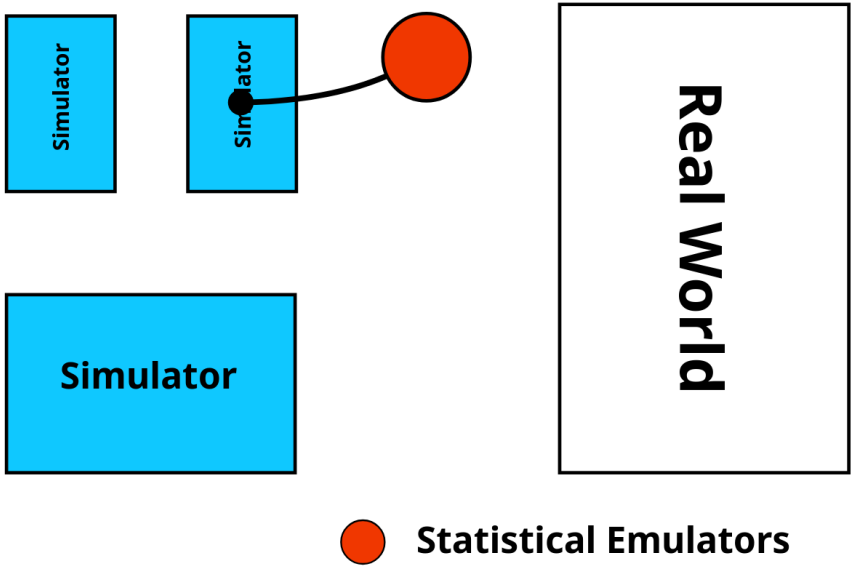
- Why is uncertainty not just desirable but necessary for learning
- Why any prediction can only be interpreted in the light of the beliefs that lead to them
- Mathematical formalism for computing with uncertainties
- Gaussian processes as a means of parametrising beliefs over functions


CO_2
Atmosphere
Carbon Cycle
Volcanoes
Fossil Fuel
Vegetation
soil
organic matter
Rivers
Surface Ocean
Marine Biota
Dissolved Organic
Carbon
Deep Ocean
Sediments

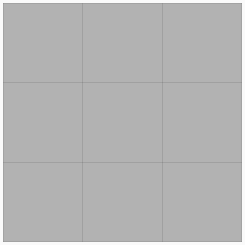
Simulator
Simulator
Simulator
Real World

Types of Simulation

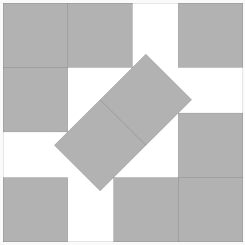
- Differential Equation Models
  - epidemiolgocial models, climate/weather models
- Discrete Event Simulations
  - turn-based
  - event-driven
- Computer Code


Simulator
Simulator
Simulator
Real World
Statistical Emulators

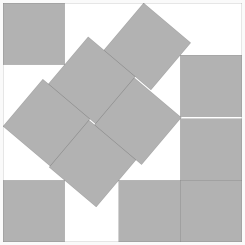
Packing Problem [1]



$s = 3$



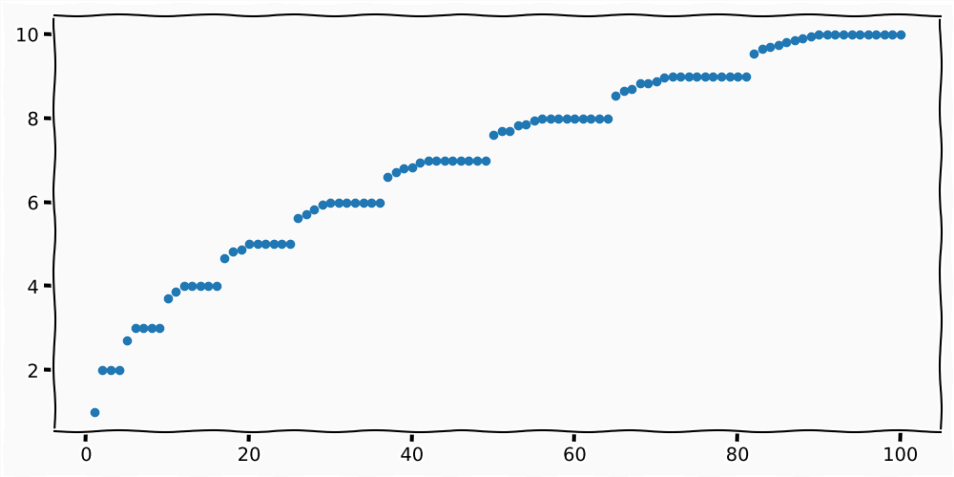
$s = 3 + \frac{1}{2}\sqrt{2}$



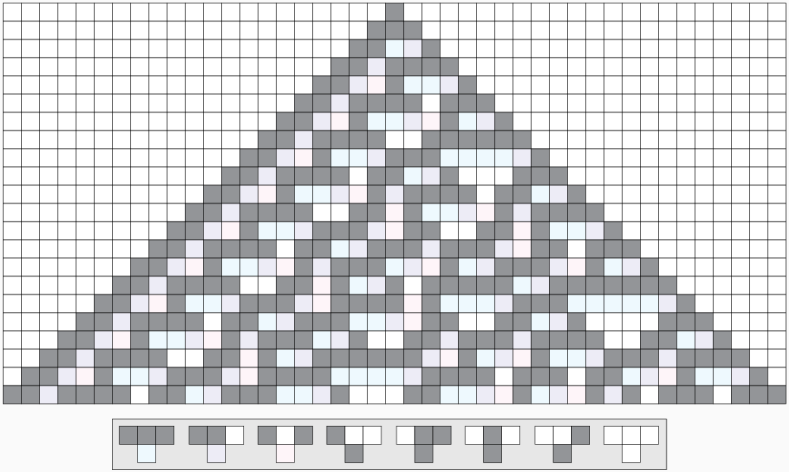
$$s^8 - 20s^7 + 178s^6 - 842s^5 + 1923s^4 - 496s^3 - 6754s^2 + 12420s - 6865 = 0$$

[1] apa

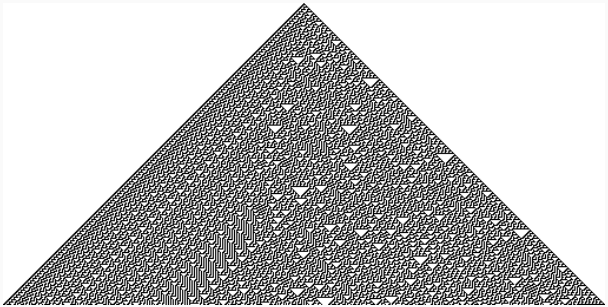
Data Regularity



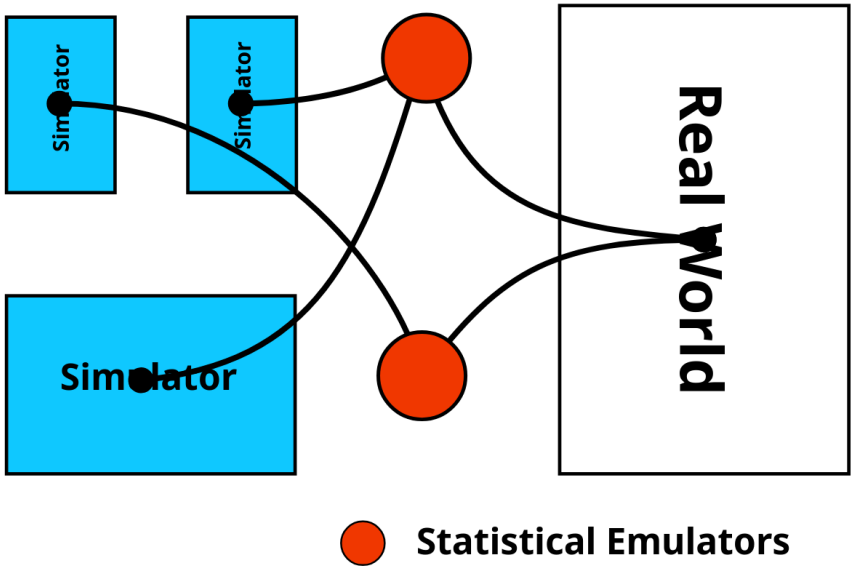
Data is Compute



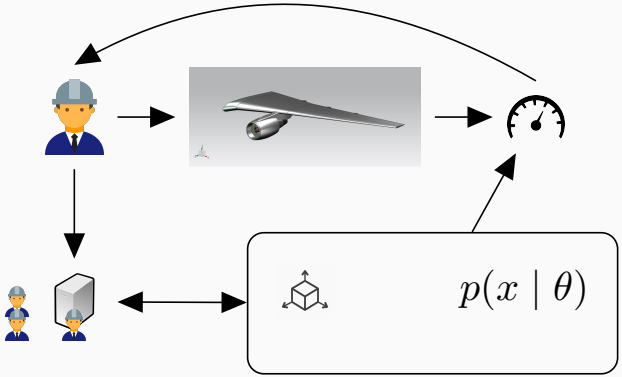
Data is Compute



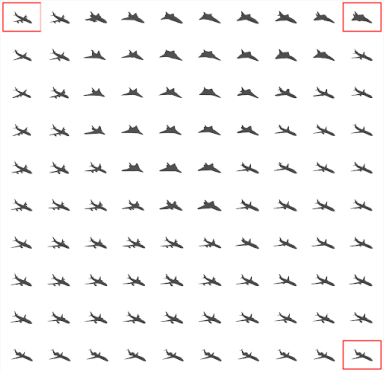
Current Pattern	$1,1,1$	$1,1,0$	$1,0,1$	$1,0,0$	$0,1,1$	$0,1,0$	$0,0,1$	$0,0,0$
New State	0	0	0	1	1	1	1	0

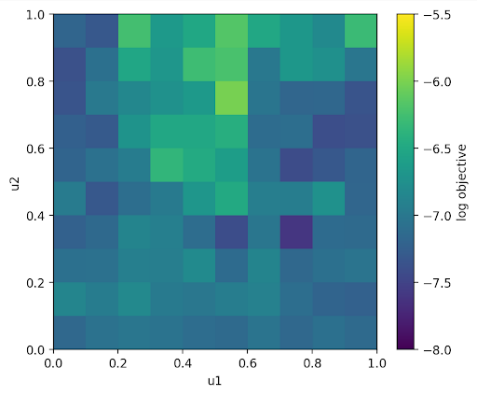

Simulator
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Simulator
Real World
Statistical Emulators

Boeing Example



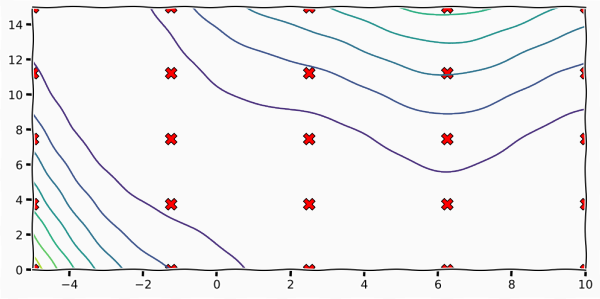
Boeing Example





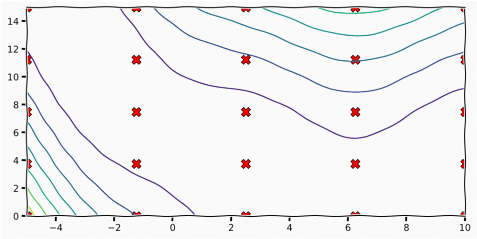
Statistical Emulation

Branin Function



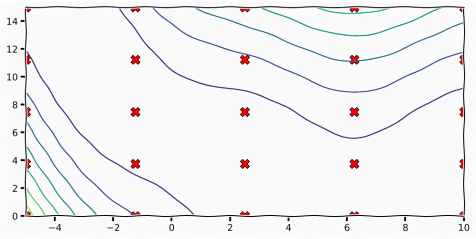
$$f(\mathbf{x}) = a(x_2 - bx_1^2 + cx_1 - r)^2 + s(1 - t\cos(x_1)) + s$$

Branin Function



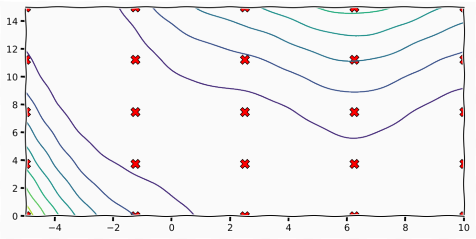
$$p(U) \sim \mathbb{U}[(-5, 10) \times (0, 15)]$$
$$\mathbb{E}[f(U)] = ?$$

Branin Function



$$\mathbb{E}[f(U)] \approx \frac{1}{n}\sum_{i}^{N} f(\mathbf{x}_i) = 73.5$$

Branin Function

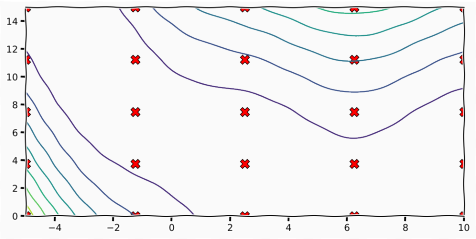


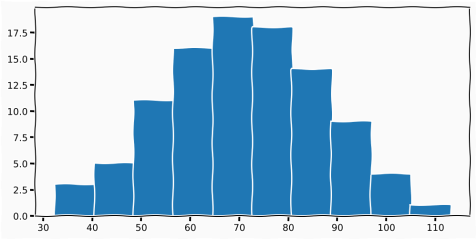
Fit a GP surrogate

$$k(\mathbf{x}_i, \mathbf{x}_j) = k_{\text{Exp}}(\mathbf{x}_i, \mathbf{x}_j) + k_{\text{White}}(\mathbf{x}_i, \mathbf{x}_j) + k_{Bias}$$

$$\mathbb{E}[f(U)] \approx \mathbb{E}[p(\mathbf{f} \mid \mathbf{Y}, \mathbf{X}, \mathbf{X}_{\text{samples}})] = 70.25$$

Branin Function






contex
context
ontex

Questions that we want to answer with Emulation

- Where to query data next?

Questions that we want to answer with Emulation

- Where to query data next?
- What to sample next?

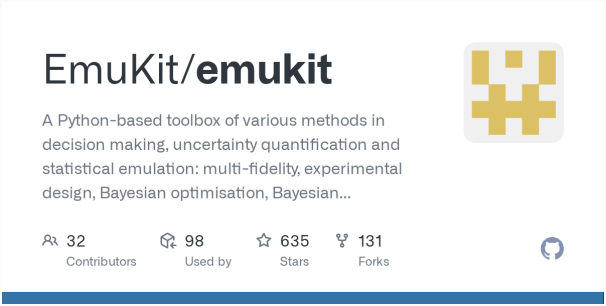
Questions that we want to answer with Emulation

- Where to query data next?
- What to sample next?
- How certain is my current estimate?

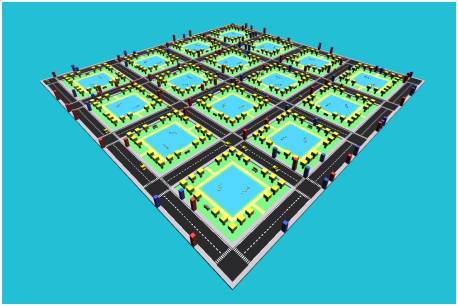
Questions that we want to answer with Emulation

- Where to query data next?
- What to sample next?
- How certain is my current estimate?
- What is the most efficient way to reduce my uncertainty?

EmuKit



https://emukit.github.io/ https://proceedings.scipy.org/articles/gerudo-f2bc6f59-009



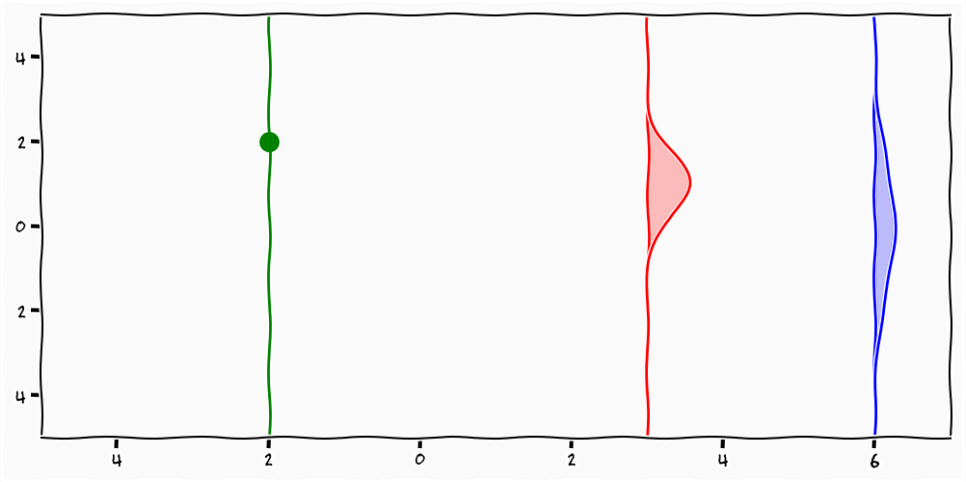
https://emukit.github.io/emukit-playground/

Gaussian Process Emulators

$$f \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

$$\begin{aligned} \mu(\mathbf{x}) &= \mathbb{E}[f(\mathbf{x})] \\ \operatorname{cov}(f(\mathbf{x}), f(\mathbf{x}')) &= \mathbb{E}\left[(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})])\,(f(\mathbf{x}') - \mathbb{E}[f(\mathbf{x}')])\right] \end{aligned}$$

Gaussian Process: Sample Paths



What is a Kernel: Definition

$$\operatorname{cov}\left(f(\mathbf{x}), f(\mathbf{x}')\right) = \mathbb{E}\left[\left(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]\right)\left(f(\mathbf{x}') - \mathbb{E}[f(\mathbf{x}')]\right)\right]$$

- Positive

$$k(\mathbf{x}, \mathbf{x}) = \operatorname{cov}(f(\mathbf{x}), f(\mathbf{x})) = \operatorname{var}(f(\mathbf{x})) \geq 0$$

- Symmetric

$$k(\mathbf{x}, \mathbf{x}') = \operatorname{cov}(f(\mathbf{x}), f(\mathbf{x}')) = \operatorname{cov}(f(\mathbf{x}'), f(\mathbf{x}))$$

$$T = \sum_{i=1}^{n} a_i f(\mathbf{x}_i)$$

$$\operatorname{var}(T) = \operatorname{cov}\left(\sum_{i=1}^{n} a_i f(\mathbf{x}_i), \sum_{i=1}^{n} a_j f(\mathbf{x}_j)\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \operatorname{cov}\left(f(\mathbf{x}_i), f(\mathbf{x}_j)\right)$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0$$

Definition (PSD)

A positive semi-definite function satisfies the above inequality for all $n \in \mathbb{N}$, $\mathbf{x}_i \in \mathcal{D}$ and $a_i \in \mathbb{R}$

Theorem (Loeve)

$k(\cdot, \cdot)$ is a covariance of a GP $\Leftrightarrow$ $k(\cdot, \cdot)$ is a positive semi-definite function

Squared Exponential $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{(\mathbf{x}-\mathbf{x}')^2}{2\ell^2}\right)$

Matern 5/2 $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{5}|\mathbf{x}-\mathbf{x}'|}{\ell} + \frac{5|\mathbf{x}-\mathbf{x}'|^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}|\mathbf{x}-\mathbf{x}'|}{\ell}\right)$

Matern 3/2 $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}|\mathbf{x}-\mathbf{x}'|}{\ell} +\right) \exp\left(-\frac{\sqrt{3}|\mathbf{x}-\mathbf{x}'|}{\ell}\right)$

Exponential $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{|\mathbf{x}-\mathbf{x}'|}{\ell}\right)$

Rational Quadratic $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{(\mathbf{x}-\mathbf{x}')^2}{2\alpha\ell^2}\right)^{-\alpha}$

Periodic $k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{2\sin^2(\pi\frac{|\mathbf{x}-\mathbf{x}'|}{p})}{\ell^2}\right)$

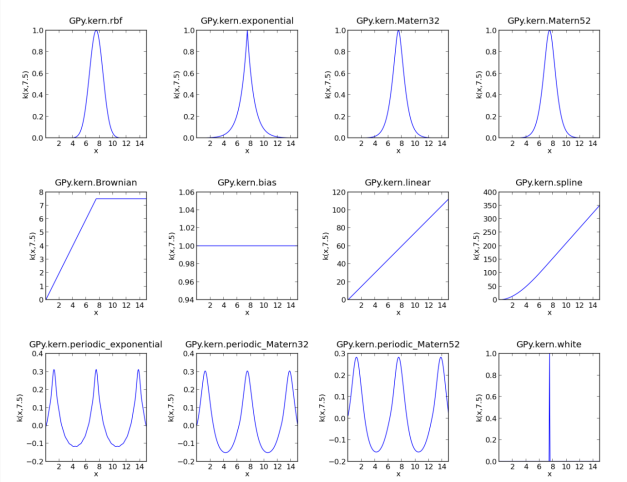
Brownian $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \min(\mathbf{x}, \mathbf{x}')$

White $k(\mathbf{x}', \mathbf{x}) = \sigma^2 \delta_{\mathbf{x},\mathbf{x}'}$

Constant $k(\mathbf{x}, \mathbf{x}') = \sigma^2$

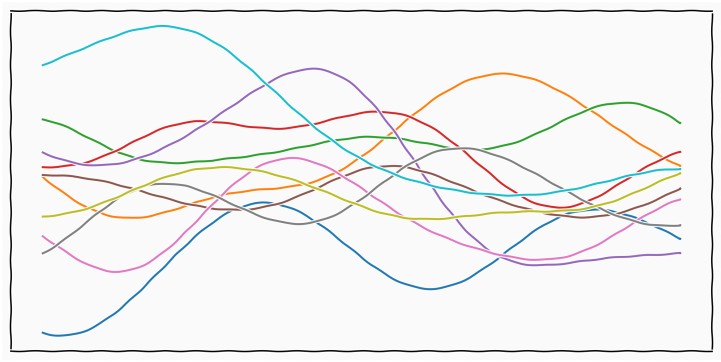
Linear $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \mathbf{x}^{\mathrm{T}} \mathbf{x}'$

Response [2]



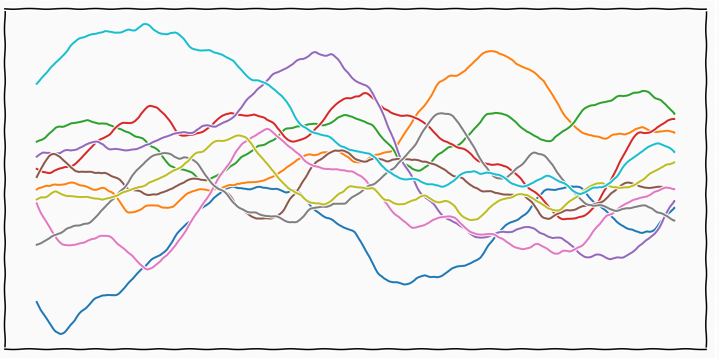
[2] Taken from Nicola Durrande

Sample Paths: Exponentiated Quadratic



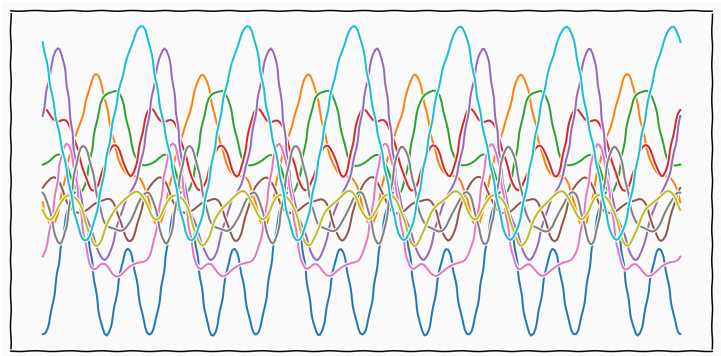
$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\ell^2}\right)$$

Sample Paths: Matern $3/2$



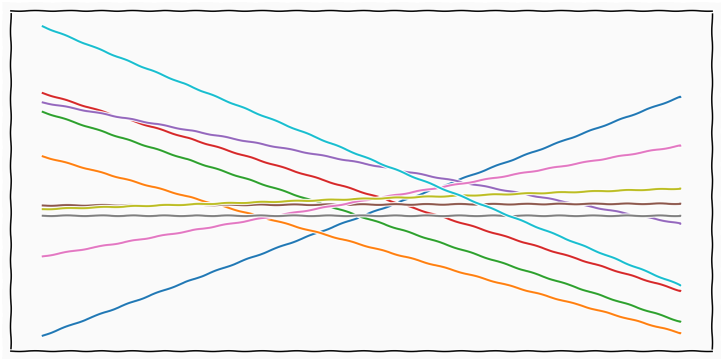
$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}|\mathbf{x} - \mathbf{x}'|}{\ell} + \right) \exp\left(-\frac{\sqrt{3}|\mathbf{x} - \mathbf{x}'|}{\ell}\right)$$

Sample Paths: Periodic



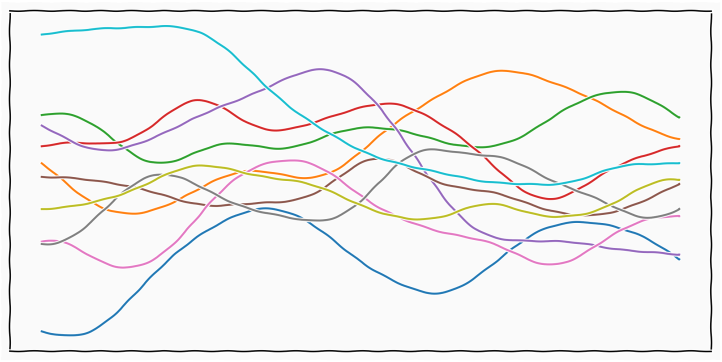
$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{2\sin^2(\pi\frac{|\mathbf{x}-\mathbf{x}'|}{p})}{\ell^2}\right)$$

Sample Paths: Linear



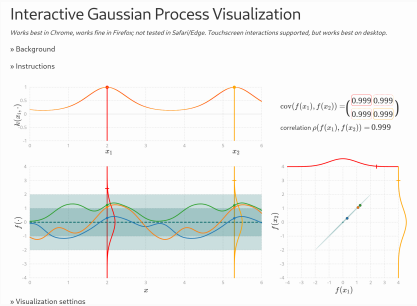
$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\ell^2}\right)$$

Sample Paths: Rational Quadratic



$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2\alpha\ell^2}\right)^{-\alpha}$$

Demo of Sample Paths



https://www.infinitecuriosity.org/vizgp/

Theorem (Bochner)

A continous stationary function $k(x, y) = \tilde{k}(|x - y|)$ is positive definite if and only if $\tilde{k}(\cdot)$ is the Fourier transform of a finite positive measure

$$\tilde{k}(t) = \int_{\mathbb{R}} e^{-i\omega t} d\mu(\omega)$$

Bochners Theorem: Example

- Positive Measure

$$\mu(\omega) = \begin{cases} 1 & -1 < \omega < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Fourier Transform

$$\tilde{k}(t) = \frac{\sin(t)}{t}$$

- Valid kernel

$$k(x, y) = \frac{\sin(x - y)}{x - y}$$

Bochners Theorem: Example

- The Gaussian is the Fourier transform of itself
- Matern kernels are the Fourier transform of

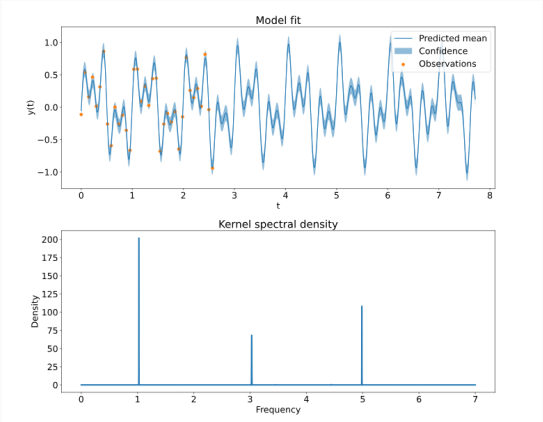
$$\mu(\omega) = \frac{1}{(1+\omega^2)^p}$$

Spectral Density

$$k(t) = \int_{\mathbb{R}} e^{-i\omega t} \mathrm{d}\mu(\omega) = \int_{\mathbb{R}} S(\omega) e^{-i\omega t} \mathrm{d}\omega$$

$$S(\omega) = \int k(t) e^{-i\omega t} \mathrm{d}t$$

Spectral Mixture Kernel [3][4]



[3]spectral mixture blog[[https://sebastiancallh.github.io/post/spectral-mixture-kernels/][]]

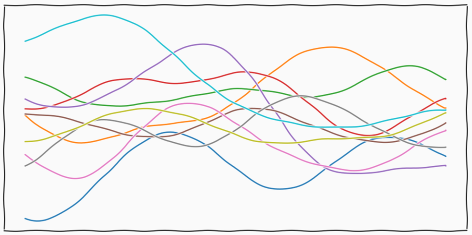
[4]Wilson et al., 2013.

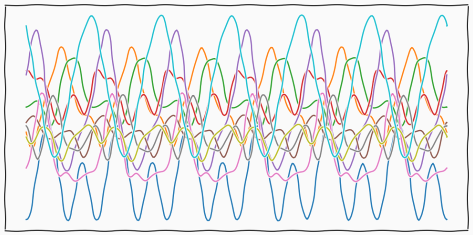
Combining Kernels (Bishop, 2006)

$$
\begin{aligned}
k(x, x') &= c k_1(x, x') \\
k(x, x') &= f(x) k_1(x, x') f(x') \\
k(x, x') &= q(k_1(x, x') \\
k(x, x') &= \exp(k_1(x, x')) \\
k(x, x') &= k_1(x, x') + k_2(x, x') \\
k(x, x') &= k_1(x, x') k_2(x, x') \\
k(x, x') &= k_3(\phi(x), \phi(x')) \\
k(x, x') &= x^{\mathrm{T}} \mathbf{A} x' \\
k(x, x') &= k_a(x_a, x'_a) + k_b(x_b, x'_b) \\
k(x, x') &= k_a(x_a, x'_a) k_b(x_b, x'_b)
\end{aligned}
$$

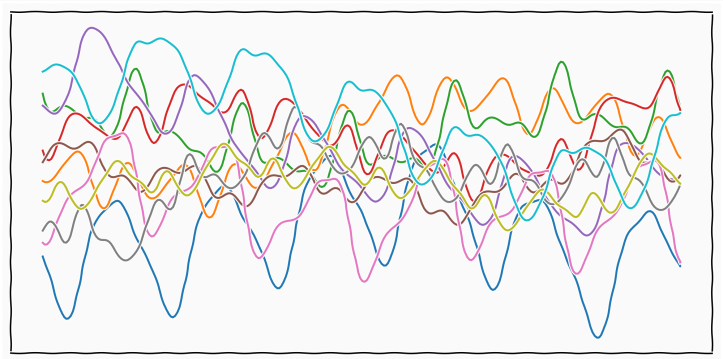
Positive definite functions are closed under certain operations

Combining Kernels



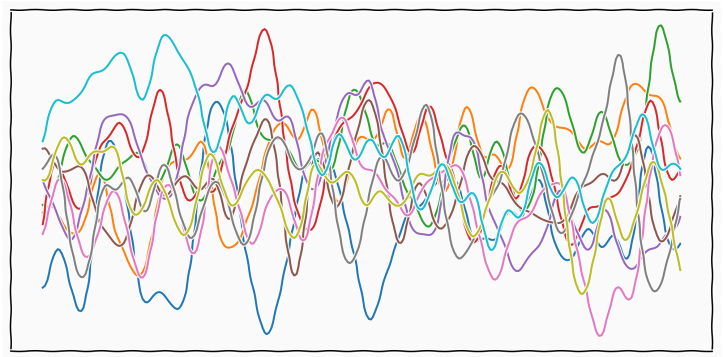


Combining Kernels: Addition



$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$$

Combining Kernels: Product



$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') \cdot k_2(\mathbf{x}, \mathbf{x}')$$

How to Choose the Kernel

To choose the co-variance we need domain knowledge

- Is the process stationary?

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- Do we expect particular patterns

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- Do we expect trends in the data?
- Do we expect particular patterns
  - periodicity?

How to Choose the Kernel

To choose the co-variance we need domain knowledge

- Is the process stationary?
- Is it differentiable, how many times, how regular are the paths?
- Do we expect trends in the data?
- Do we expect particular patterns
  - periodicity?
  - cycles?

How to Choose the Kernel

To choose the co-variance we need domain knowledge

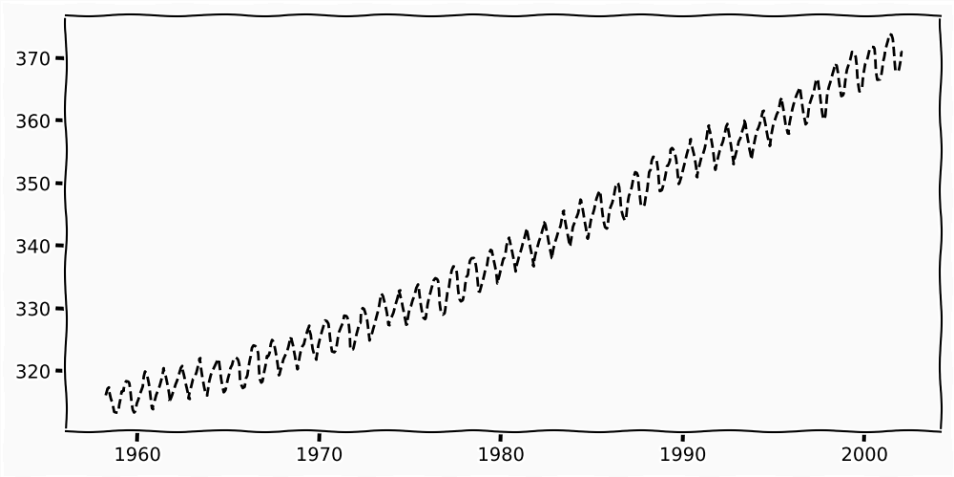
- Is the process stationary?
- Is it differentiable, how many times, how regular are the paths?
- Do we expect trends in the data?
- Do we expect particular patterns
  - periodicity?
  - cycles?
  - additivity?

How to Choose the Kernel

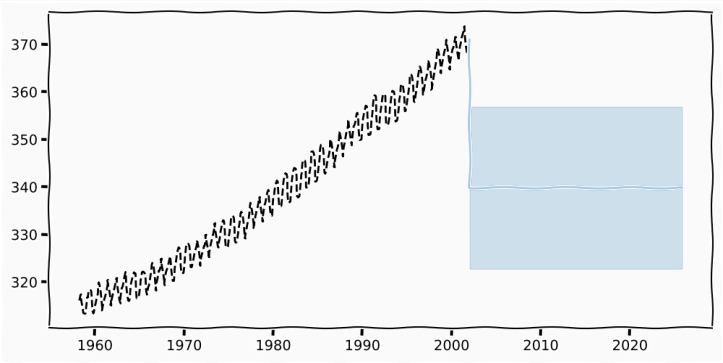
To choose the co-variance we need domain knowledge

- Is the process stationary?
- Is it differentiable, how many times, how regular are the paths?
- Do we expect trends in the data?
- Do we expect particular patterns
  - periodicity?
  - cycles?
  - additivity?
- Should we model in a transformed space?
  $x \to log(x), x \to exp(x), \ldots$

Mauna Loa (Rasmussen et al., 2006)

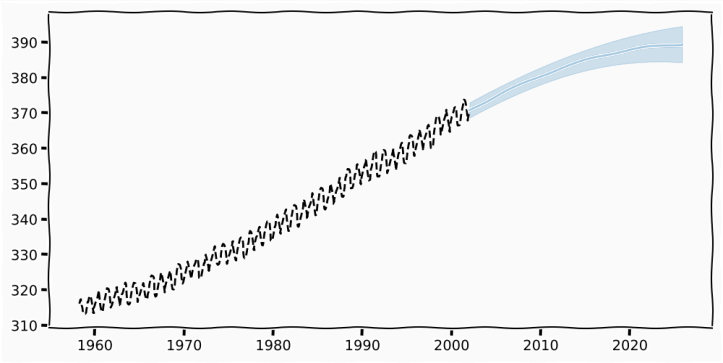


Mauna Loa (Rasmussen et al., 2006)



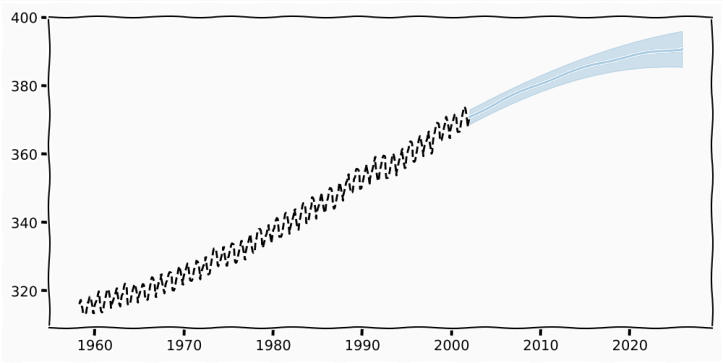
$$k(t, t') = k_{\text{SQE}}(t, t')$$

Mauna Loa (Rasmussen et al., 2006)



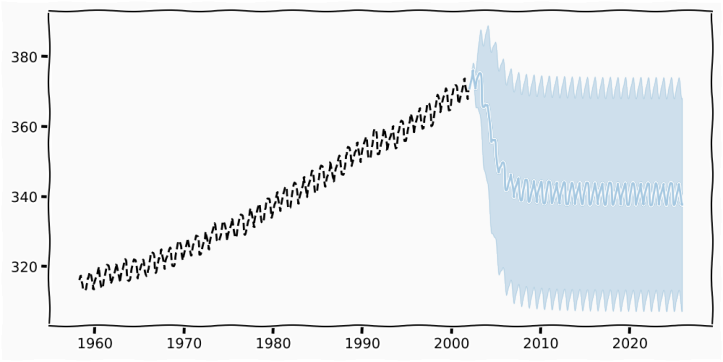
$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{White}}(t, t')$$

Mauna Loa (Rasmussen et al., 2006)



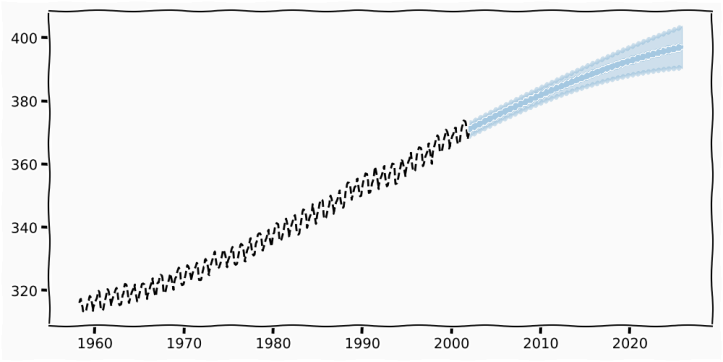
$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{White}}(t, t') + k_{\text{Periodic}}(t, t')$$

Mauna Loa (Rasmussen et al., 2006)



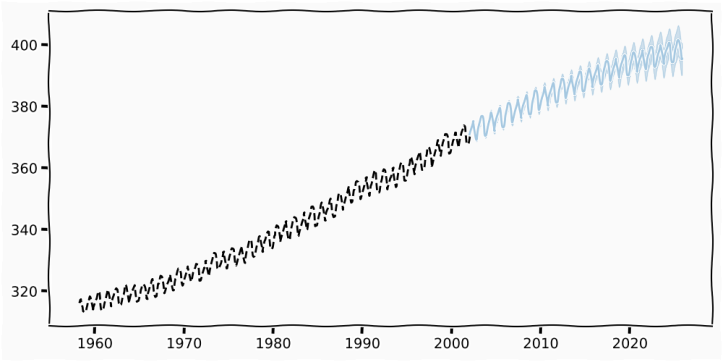
$$k(t, t') = k_{\text{SQE}}(t, t') + k_{\text{White}}(t, t') + k_{\text{Periodic}}(t, t') + k_{\text{SQE}}(t, t') + k_{\text{RQ}}(t, t')$$

Mauna Loa (Rasmussen et al., 2006)



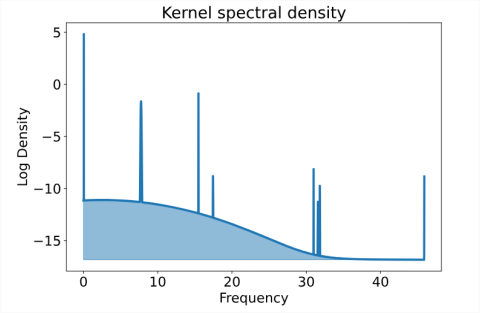
$$k(t,t') = k_{\text{SQE}}(t,t') + k_{\text{White}}(t,t') + k_{\text{Periodic}}(t,t') + k_{\text{SQE}}(t,t') + k_{\text{RQ}}(t,t') + k_{\text{lin}}(t,t')$$

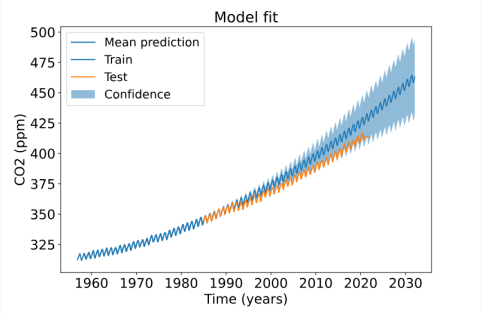
Mauna Loa (Rasmussen et al., 2006)



$$k(t,t') = k_{\text{SQE}}(t,t') + k_{\text{SQE}}(t,t') \cdot k_{\text{Periodic}}(t,t') + k_{\text{RQ}}(t,'t) + k_{\text{White}}(t,t')$$

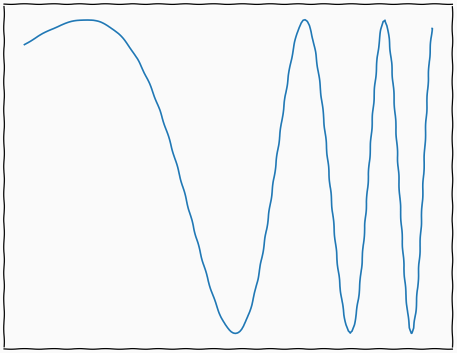
Mauna Loa [5]

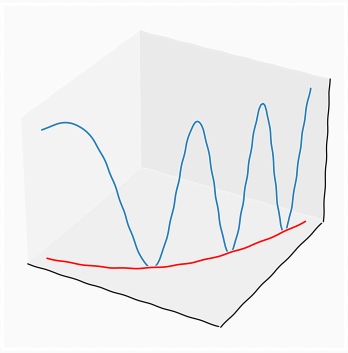




[5] spectral mixture blog

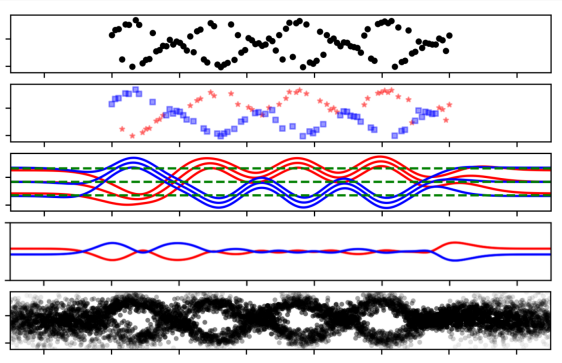
Kernels with Latent Inputs (Bodin et al., 2017)





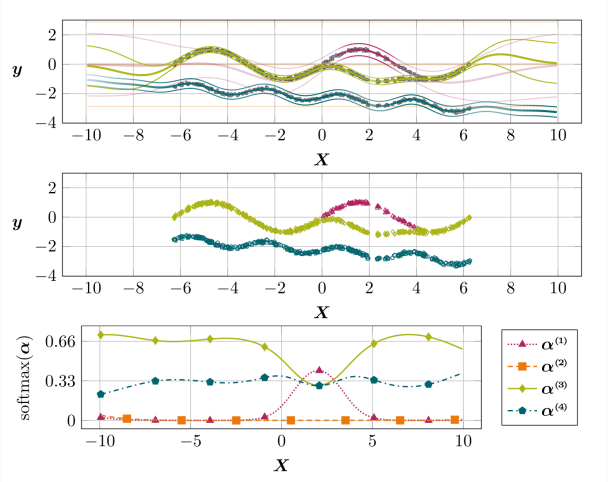
$$k\left(\left(\mathbf{x}, \mathbf{x}_l\right), \left(\mathbf{x}', \mathbf{x}_l'\right)\right)$$

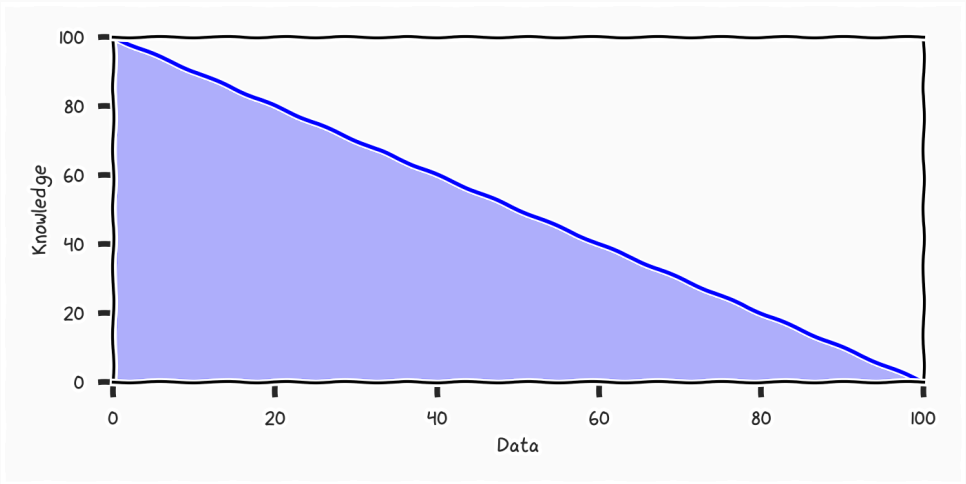
Kernels with Latent Inputs (Bodin et al., 2017)



$$k\left(\left(\mathbf{x}, \mathbf{x}_l\right), \left(\mathbf{x}', \mathbf{x}_l'\right)\right)$$

Switching Processes (Kaiser et al., 2018)




Knowledge
100
80
60
40
20
0
0
20
40
60
80
100
Data





Summary

Summary

- What is an emulator?
- How do simulations and emulation interact?
- Gaussian process emulators
  - very rich framework for inputing domain knowledge

The rest of the course

Part 1 introduction and building emulators

Part 2 what to do with emulators

eof

References

References

- Bishop, Christopher M. (2006). ***Pattern Recognition and Machine Learning (Information Science and Statistics)***. Secaucus, NJ, USA: Springer-Verlag New York, Inc.
- Bodin, Erik, Neill D. F. Campbell, and Carl Henrik Ek (2017). **"Latent Gaussian Process Regression"**. In: *CoRR*.
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- Rasmussen, Carl Edward and Christopher K I Williams (2006). *Gaussian Processes for Machine Learning (Adaptive Computation and Machine Learning)*. The MIT Press.
- Wilson, Andrew and Ryan Adams (2013). “Gaussian Process Kernels for Pattern Discovery and Extrapolation”. In: *Proceedings of the 30th International Conference on Machine Learning*. Ed. by Sanjoy Dasgupta and David McAllester. Vol. 28. Proceedings of Machine Learning Research 3. Atlanta, Georgia, USA: PMLR, pp. 1067–1075.